Competing Risks and Multi-State Models

Concepts, methods and software

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Part V

Software

Contents:

1. Nonparametric estimation and plotting (Kaplan-Meier, Aalen-Johansen)
2. Creating data set with time-dependent censoring weights
3. PL-estimator and Fine and Gray model
4. Prediction

Outline

Nonparametric estimation: Aalen-Johansen

Single event type
Aalen-Johansen estimator

Nonparametric Estimation of Subdistribution
Time-varying values; counting process format
Representation of the weights
Creation of the data set with weights
The product-limit estimator
Log-rank tests

Fine & Gray model

Prediction based on Cause-specific Hazard

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Right Censored Data

Representation Two columns: **time** since origin and **status** variable. `status=1` if event observed and `status=0` if event right censored

```
id time status
 1   7.7      1
 2   4.3      1
 3   5.6      0
 ...
```

In R via `Surv(time=time, event=status)`

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```
id time status
 1  7.7      1
 2  4.3      1
 3  5.6      0
 ...
```

In R via `Surv(time=time, event=status)`

Kaplan-Meier Main function: `survfit.formula`

```
survfit(Surv(time,status)~1, data=...)
```

Four example individuals

`aidssi` data set (available in `mstate` package and in Stata)

patnr	time	status	cause	ccr5
14	5.054	0	event-free	WW
3	2.234	1	AIDS	WW
15	10.196	1	AIDS	WM
8	8.605	2	SI	WW

Four example individuals

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Kaplan-Meier both event types combined:

```
KM.overall <- survfit(Surv(time,status!=0)~1,
                                          data=aidssi)
```

Note: `event` argument can be a logical expression (`status!=0`)

Four example individuals

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patnr	**time**	**status**	**cause**	**ccr5**
14	5.054	0	event-free	WW
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```
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                                              data=aidssi)
```

Note: `event` argument can be a logical expression (`status!=0`)

Kaplan-Meier per value of CCR5:

```
KM.ccr5 <- survfit(Surv(time,status!=0)~ccr5,
                                              data=aidssi)
```

Numerical summary

Basic summary:

```
KM.overall

       n events median 0.95LCL 0.95UCL
[1,] 329    222    7.3    6.44    8.36
```

More detailed summary:

```
summary(KM.overall)

   time n.risk n.event survival std.err lower 95% CI upper 95% CI
  0.112    329       1    0.997 0.00303        0.991        1.000
  0.137    328       1    0.994 0.00429        0.986        1.000
  0.474    325       1    0.991 0.00525        0.981        1.000
      .      .       .        .       .            .            .
 12.936     41       1    0.217 0.02604        0.171        0.274
 13.361     22       1    0.207 0.02665        0.161        0.266
 13.936      1       1    0.000     NaN           NA           NA
```

Survival at 12 years obtained via

```
summary(KM.overall, time=12)
```

In R

Uses the function `survfit.formula`, which calls `survfitKM`

```
survfitKM
 function (x, y, weights = rep(1, length(x)),
     stype = 1, ctype = 1,
     se.fit = TRUE, conf.int = 0.95,
     conf.type = c("log", "log-log", "plain", "none",
                                               "logit", "arcsin"),
     conf.lower = c("usual", "peto", "modified"),
     start.time, id, cluster, robust,
     influence = FALSE, type, entry = FALSE, time0 = FALSE)
```

Plotting Kaplan-Meier survival curves

`plot.survfit` for first plot:

```
function (x, conf.int,
  mark.time = FALSE, pch = 3, col = 1, lty = 1, lwd = 1, cex =
  log = FALSE, xscale = 1, yscale = 1, xlim, ylim,
  xmax, fun, xlab = "", ylab = "", xaxs = "r",
  conf.times, conf.cap = 0.005, conf.offset = 0.012,
conf.type = c("log", "log-log", "plain", "logit", "arcsin", "
  mark.col, noplot = "(s0)", cumhaz = FALSE, firstx, ymin,
  cumprob = FALSE, ...)
```

`fun="event"` plots cumulative incidence (i.e. upwards from 0):

```
plot(KM.overall, fun="event")
```

`lines.survfit` adds curves

`ggsurvfit` package

- Extension of `ggplot2`.
  See `https://www.danieldsjoberg.com/ggsurvfit`
- Allows for number-at-risk table, automatic legend
- Basic survival function via

  ```
  ggsurvfit(survfit2(Surv(time,status!=0)~ccr5,
                                   data=aidssi))
  ```

  Use `survfit2` for nicer labeling of groups
- `type="risk"` plots cumulative incidence
- CIs not plotted by default; use
  `+ add_confidence_interval()`

survminer package

- Extension of ggplot2.
  See https://rpkgs.datanovia.com/survminer/index.html
- Allows for number-at-risk table, automatic legend
- Basic survival function via

```
ggsurvplot(survfit(Surv(time,status!=0)~ccr5,
                                          data=aidssi))
```

- fun="event" plots cumulative incidence

Four example individuals

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Kaplan-Meier both event types combined:

```
KM.overall <- survfit(Surv(time,status!=0)~1,
                                          data=aidssi)
```

Note: `event` argument can be a logical expression (`status!=0`)

Kaplan-Meier per value of CCR5:

```
KM.ccr5 <- survfit(Surv(time,status!=0)~ccr5,
                                          data=aidssi)
```

Results Cox model for combined end point

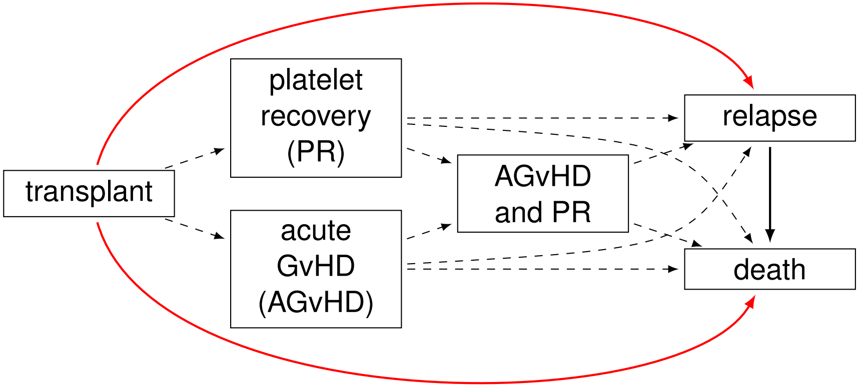
```
coxph(formula = Surv(time, status != 0) ~ ccr5,
                                    data = aidssi)
```

Result:

```
         coef  exp(coef)  se(coef)     z       p
ccr5WM -0.701     0.496    0.186 -3.77 0.00016

Likelihood ratio test=16.5  on 1 df, p=4.95e-05  n= 324,
                       number of events= 220
   (5 observations deleted due to missingness)
```

Computer practical



1. A first look at the data Have a look at the data explanation:
   `help(ebmt1)`
   Inspect the data.
   Summarize all the variables. What do you observe?
2. Estimation of overall survival Compute the Kaplan-Meier estimator for survival. Give a numerical summary and plot the estimate using the `survival`, `ggsurvfit` and `survminer` packages.
3. A Cox model for overall survival Compute and plot the Kaplan-Meier estimator for survival by value of **score**, using the same three packages as in 2. Fit a Cox model for the effect of the EBMT score on survival.

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Software to compute Aalen-Johansen estimator

- Stata: `stcompet` command
- SAS: macros `%CUMINCID` or `%CIF`
- R, some options:
  - standard `survival` package
  - `cmprsk` or `prodlim` package
  - any package for multi-state models, e.g. `etm`, `mstate`, `msSurv`

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  - any package for multi-state models, e.g. `etm`, `mstate`, `msSurv`

Aalen-Johansen estimator

Event column needs to be a *factor*:

```
aidssi$cause <- relevel(aidssi$cause, ref="event-free")
csiSurv <- survfit(Surv(time, event=cause)~1,
                                              data=aidssi)
```

Summary of estimate:

```
summ <- summary(csiSurv, times=seq(2,8,by=2))

summ
 time n.risk n.event Pr((s0)) Pr(AIDS) Pr(SI)
    2    300      15    0.953  0.00632 0.0404
    4    240      52    0.786  0.10322 0.1112
    6    170      54    0.603  0.17070 0.2259
    8    122      41    0.454  0.25430 0.2916
```

Confidence intervals: `summ$lower` and `summ$upper`

Plotting

- `survival`: use same `plot` function
  - specific event types selected via `csiSurv["AIDS"]` etc.
  - default curve shown from 0 upwards `fun="event"`
- `ggsurvfit`: use `ggcuminc` function
  - specific event types selected via `outcome` argument
  - alternate and stacked display format not implemented

```
ggcuminc(csiSurv, outcome=c("AIDS","SI"))
```

- `survminer`: use `ggcompetingrisks` function
  - less flexible than `ggcuminc` in `ggsurvfit`

5. Estimation of cause-specific cumulative incidence Compute the Aalen-Johansen estimator for relapse and relapse-free mortality. What is the probability, with 95% confidence intervals (on the default log scale), to have a relapse within one year and within five years.
   Compare the estimates at 1 and 5 years with the relapse-free survival distribution, i.e. the distribution of time to remain free of both event types.
6. Some plots (a) Plot the estimated cause-specific cumulative incidence for each end point using the overlaid display format. Plot the 95% confidence intervals as well.
   (b) Plot the estimated cause-specific cumulative incidence using the stacked format (without the confidence intervals).
   (c) Plot the cause-specific cumulative incidence estimates for each end point using the alternate display format.
7. Regression on the cause-specific hazards Fit a proportional hazards model for relapse and survival as competing risks.

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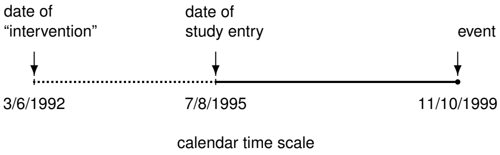
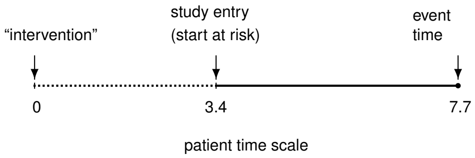
Left truncated data: structure

- In calendar time scale



- In patient time scale



Left truncated data: analysis and data representation

Individuals only contribute while they are in follow-up

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- Data: extra column, describing entry time in risk set

id	entry time	event time	status
1	0.0	4.3	1
2	0.0	5.6	0
3	3.4	7.7	1

- In R: `Surv(entry.time, event.time, status)`

Left truncated data: analysis and data representation

Individuals only contribute while they are in follow-up

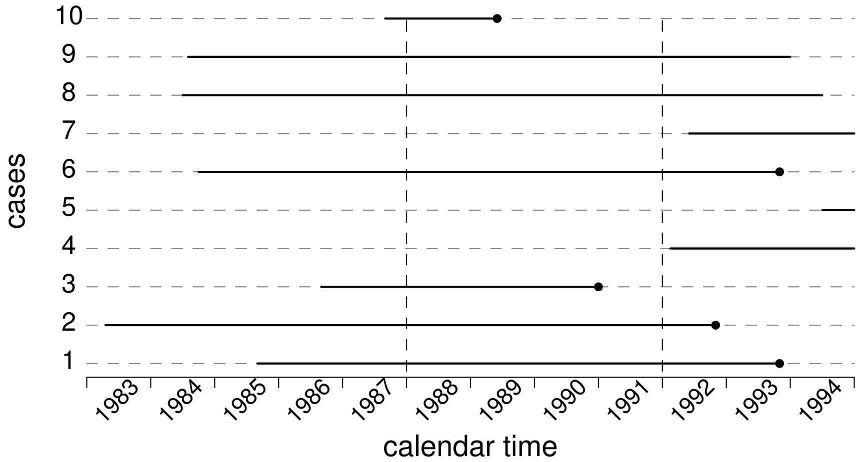
- Data: extra column, describing entry time in risk set

```
id      entry time      event time      status
 1             0.0             4.3           1
 2             0.0             5.6           0
 3             3.4             7.7           1
```

- In R: `Surv(entry.time, event.time, status)`
- Note: log-rank test function `survdiff` does not work with left-truncated data; use equivalence with score test from Cox model

```
> coxph(
  Surv(entry.time, event.time, status) ~ ccr5,
                                  ...)$score
```

Example time-varying covariable



Data format

- start time, stop time and event status at stop time, plus covariable value in interval.
  Represented by three rows:

id	Tstart	Tstop	status	cal.period
1	0	2.3	0	< 1988
1	2.3	6.3	0	1988-1991
1	6.3	8.2	1	> 1991

  Enters in risk set "1988-1991" ("$>$ 1991") after 2.3 (6.3) years

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- R: `Surv(Tstart, Tstop, status) ~ cal.period`

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  Enters in risk set “1988-1991” (“$>$ 1991”) after 2.3 (6.3) years
- R: `Surv(Tstart, Tstop, status) ~ cal.period`
- Similar for discontinuous risk intervals and for time updated weights

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Example data

Three individuals from data set

id	entry.time	event.time	event.type
1	0.25460	0.63644	0
2	0.00000	0.64358	1
3	0.08005	0.25615	2

General product-limit estimator of F_k

$$\widehat{\overline{F_k}}^{\mathrm{PL}}(t) = \prod_{i:t_{(i)}\le t}\left\{1 - \widehat{h_k}(t_{(i)})\right\}$$

$$\widehat{h_k}(t_{(i)}) = \frac{d_k(t_{(i)})}{r^*(t_{(i)})}$$

Weights: contribution $\omega_l(t_{(i)})$ of individual l to $r^*(t_{(i)})$ is

- censored or event of type k before $t_{(i)}$: 0
- still at risk at $t_{(i)}$: 1
- competing event at $t_{(j)}$ before $t_{(i)}$: weight

$$\frac{\widehat{\overline{\Gamma}}(t_{(i)}-)}{\widehat{\overline{\Gamma}}(t_{(j)}-)} \times \frac{\widehat{\Phi}(t_{(i)}-)}{\widehat{\Phi}(t_{(j)}-)}$$

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Three individuals from data set

id	entry.time	event.time	event.type
1	0.25460	0.63644	0
2	0.00000	0.64358	1
3	0.08005	0.25615	2

$\Longrightarrow$

id	Tstart	Tstop	status	weight.cens	weight.trunc	count	failcode
1	0.25460	0.63644	0	1.00000	1.00000	1	1
2	0.00000	0.64358	1	1.00000	1.00000	1	1
3	0.08005	0.25615	2	1.00000	1.00000	1	1
3	0.25615	0.31778	2	1.00000	1.02941	2	1
3	0.31778	0.37693	2	1.00000	1.02941	3	1
3	0.37693	0.38928	2	1.00000	1.02941	4	1
3	0.38928	0.46029	2	1.00000	1.02941	5	1
3	0.46029	0.50979	2	1.00000	1.02941	6	1
3	0.50979	0.64358	2	0.67849	1.07230	7	1
3	0.64358	0.64724	2	0.67849	1.07230	8	1
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

Events of type 1 observed at 0.31778 0.37693 0.38928 0.46029 0.50979 0.64358 0.64724

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Software

Stata	`stcrprep` command
SAS	`%PSHREG` macro
R	`finegray` function in `survival` package
	`crprep` function in `mstate` package

R: create data set with weights

patnr	time	status	cause	ccr5
14	5.054	0	event-free	WW
3	2.234	1	AIDS	WW
15	10.196	1	AIDS	WM
8	8.605	2	SI	WW

```
finegray(Surv(time,cause)∼., data=aidssi, etype="AIDS")
```

Note: `event-free` first level in `factor` specification

```
crprep(Tstop="time", status="status", data=aidssi,
                            trans=1, cens=0, keep="ccr5")
aidssi.w <- crprep(Tstop="time", status="cause",
            data=aidssi, trans="AIDS", cens="event-free",
                                                 keep="ccr5")
```

Both event types at once via `trans=c("AIDS","SI")`

Resulting data set (`crprep`)

patnr	time	status	cause	ccr5
14	5.054	0	event-free	WW
3	2.234	1	AIDS	WW
15	10.196	1	AIDS	WM
8	8.605	2	SI	WW

	id	Tstart	Tstop	status	weight.cens	ccr5	count	failcode
3	3	0.000	2.234	AIDS	1.000	WW	1	AIDS
17	8	0.000	8.605	SI	1.000	WW	1	AIDS
18	8	8.605	8.638	SI	0.991	WW	2	AIDS
19	8	8.638	8.755	SI	0.982	WW	3	AIDS
.	.	.	.	.	.	.		
78	14	0.000	5.054	event-free	1.000	WW	1	AIDS
79	15	0.000	10.196	AIDS	1.000	WM	1	AIDS

`finegray`:

- different column names **patnr, fgstart, fgstop, fgwt**
- **status** converted to numeric; extra **fgstatus** 0-1 column
- no **count** and **failcode** column

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PL-form: Kaplan-Meier with probability weights

Stata Specify `pweights` option in `stset` command; use standard `sts` command

SAS `PROC LIFEREG`

R `weights` argument in `survfit` function (`survival` package)

```
survfit(Surv(Tstart,Tstop,status=="AIDS")~1,
                       data=aidssi.w, weights=weight.cens)
```

PL-form: Kaplan-Meier with probability weights

Stata Specify `pweights` option in `stset` command; use standard `sts` command

SAS `PROC LIFEREG`

R `weights` argument in `survfit` function (`survival` package)

```
survfit(Surv(Tstart,Tstop,status=="AIDS")~1,
                  data=aidssi.w, weights=weight.cens)
```

- Additional left truncation:

  `weights=weight.cens*weight.trunc`

Both event types at once: resulting data set

	id	Tstart	Tstop	status	weight.cens	ccr5	count	failcode
3	3	0.000	2.234	AIDS	1.000	WW	1	AIDS
17	8	0.000	8.605	SI	1.000	WW	1	AIDS
18	8	8.605	8.638	SI	0.991	WW	2	AIDS
19	8	8.638	8.755	SI	0.982	WW	3	AIDS
.	.	.	.	.	.	.	.	.
78	14	0.000	5.054	event-free	1.000	WW	1	AIDS
79	15	0.000	10.196	AIDS	1.000	WM	1	AIDS
			⋮					
2768	3	0.000	2.234	AIDS	1.000	WW	1	SI
2769	3	2.234	2.322	AIDS	0.997	WW	2	SI
2770	3	2.322	2.982	AIDS	0.993	WW	3	SI
.	.	.	.	.	.	.		
2870	8	0.000	8.605	SI	1.000	WW	1	SI
2913	14	0.000	5.054	event-free	1.000	WW	1	SI
2914	15	0.000	10.196	AIDS	1.000	WM	1	SI
2915	15	10.196	10.448	AIDS	0.974	WM	2	SI
2916	15	10.448	10.467	AIDS	0.961	WM	3	SI
.	.	.	.	.	.	.	.	.

Both event types at once: estimation

```
csiPL <- survfit(
      Surv(Tstart,Tstop,status==failcode)~failcode,
            data=aidssi.w, weights=weight.cens)
csiPL$strata
## failcode=AIDS   failcode=SI
##          310            310
summary(csiPL["failcode=SI"],times=seq(2,10,by=2))
```

and obtain

```
 time n.risk n.event survival std.err lower 95% CI upper 95% CI
    2    302      13 0.959615  0.0110     0.938344     0.981369
    4    272      22 0.888783  0.0177     0.854687     0.924239
    6    218      34 0.774112  0.0240     0.728466     0.822619
    8    190      18 0.708440  0.0265     0.658364     0.762324
   10    161      11 0.665410  0.0279     0.612936     0.722377
```

or use `csiPL[2]`

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Gray's log-rank test

- Implemented directly, using original (unweighted) data set
  SAS: `%CIF` macro
  R: `cuminc` function in `cmprsk` package
- Use score test from proportional subdistribution hazards model
  - Compute censoring weights per subgroup (e.g. via `strata` argument in `crprep` function)

```
aidssi.wCCR <- crprep(Tstop="time",
  status="cause", data=aidssi,
  trans=c("AIDS","SI"), cens="event-free",
  id="patnr", strata="ccr5")
```

  - Variance of the test statistic is different from the one suggested by Gray. Affects p-value

R example code

```
test.AIDS <- coxph(Surv(Tstart,Tstop,status=="AIDS")~ccr5,
                    data=aidssi.wCCR, weights=weight.cens,
                                      subset=failcode=="AIDS")
test.SI <- coxph(Surv(Tstart,Tstop,status=="SI")~ccr5,
                  data=aidssi.wCCR, weights=weight.cens,
                                      subset=failcode=="SI")
```

Score test statistic and p-value

```
## AIDS
c(test.AIDS$score, 1-pchisq(test.AIDS$score,1))
[1] 12.428139562  0.000422913
## SI
c(test.SI$score, 1-pchisq(test.SI$score,1))
[1] 0.007207246 0.932344471
```

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- Creation of the data set with weights
- The product-limit estimator
- Log-rank tests

Fine & Gray model

Prediction based on Cause-specific Hazard

Proportional subdistribution hazards (Fine & Gray)

- Implemented directly (using robust estimator of standard error; no left truncation)

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SAS	`PROC PHREG` procedure (SAS/STAT $\geq$ 13.1)
R	`crr` function in `cmprsk` package; `FGR` function from `riskRegression` package provides a formula interface

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- Translate to cumulative incidence: use standard code for prediction based on results of Cox model with weights

Fitting the model and obtaining predictions

```
fit.FG <- coxph(Surv(Tstart,Tstop,status==failcode) ~
                ccr.comb + strata(failcode), data=aidssi.w,
                   robust=FALSE, weights=weight.cens)
```

Predict for four possible combinations

```
indivs <- data.frame(
        ccr.comb=factor(c("WW","WW","WM.AIDS","WM.SI")),
                   failcode=c("AIDS","SI","AIDS","SI"))
pred.sdh <- survfit(fit.FG, newdata=indivs)
```

Results per combination, e.g. for SI and `WW`:

```
pred.sdh[2]
```

8. Estimate weights and compute PL-estimates. Compute the data set with weights for relapse. Include **type** and **type** and the covariables **score** and **age**. Inspect the data. Compute and plot the PL-form for relapse and add AJ-form. Compare estimates and 95% CIs.
9. Estimate per **score** value, based on PL-estimate Compute the weights and the PL-estimates. Plot the results, using the overlaid, stacked and alternate formats. Compute weights by value of **score**? Estimate and plot the time-to-censoring distribution by **score**.
10. Log-rank tests Run and compare log-rank tests
11. Impact of EBMT score on subdistribution hazards Compare the parameter estimates with the ones from the proportional cause-specific hazards model. Can you explain the difference?

Outline

Nonparametric estimation: Aalen-Johansen

- Single event type
- Aalen-Johansen estimator

Nonparametric Estimation of Subdistribution

- Time-varying values; counting process format
- Representation of the weights
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Fine & Gray model

Prediction based on Cause-specific Hazard

From cause-specific hazard regression to cumulative incidence

Stata	`stpm2` for parametric hazard model; `stpm2cif` translates to cumulative incidence
SAS	Macros `CumInc.sas` and `CumIncV.sas`
R	`mstate` package

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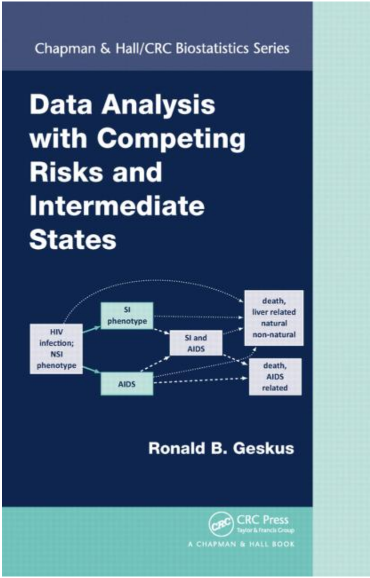

THANKS!